BhoomiSetu

SIH26016 — Project Overview & Development Blueprint

Purpose: A practical guide for a six-member first-year B.Tech team to understand the project, define the MVP, divide the work, and prepare an SIH-ready prototype.

Important note: This is a project-planning interpretation of the problem discussed in this chat. Verify the exact SIH26016 wording and requirements on the official SIH 2026 portal before final submission.

1. Project in One Minute

BhoomiSetu is a centralized digital platform for managing and monitoring the land-acquisition lifecycle. It brings projects, land parcels, landowners, documents, verification, compensation, approvals, timelines and alerts into one system.

Simple flow: Project creation → land identification → parcel/owner records → document submission → verification → valuation → compensation → approval/payment → final acquisition → project completion.

2. Problem We Are Solving

- Land-acquisition work can involve multiple departments, stakeholders, documents and approval stages.
- Officers need a single place to monitor project progress and identify pending work.
- Landowners need clearer visibility into the status of their land, documents and compensation.
- Management needs dashboards and reports to identify bottlenecks and delayed projects.
- A useful digital system should maintain an auditable history of important actions.

3. Proposed Solution

Build three connected experiences:

- **Government Officer Portal:** create/manage projects, parcels, documents, verification, compensation and reports.
- **Landowner Portal:** view own land, acquisition progress, document status, compensation and notifications.
- **Admin Portal:** manage users/departments, monitor system activity and view audit logs.

4. Core Modules

Module	Purpose	Priority
Authentication & Roles	Login and role-based access for Admin, Officer and Landowner.	MUST
Project Management	Create projects, locations, land requirements, dates and status.	MUST
Land Parcels	Maintain parcel, owner, survey, area and acquisition information.	MUST
GIS Map	Display parcels/projects with status indicators.	MUST
Documents	Upload, review, verify and track required documents.	MUST
Compensation	Track calculation, approval, processing and payment status.	MUST
Landowner Portal	Give landowners a simple view of their land and progress.	MUST
Notifications	Show document, verification, compensation and action alerts.	IMPORTANT
Delay/Risk Monitor	Flag stages pending beyond expected timelines.	IMPORTANT

Module	Purpose	Priority
Analytics & Reports	Show project, parcel, compensation and delay statistics.	IMPORTANT
Audit Logs	Record who performed important actions and when.	IMPORTANT
OCR	Extract fields from uploaded documents; prototype can be simulated initially.	ADVANCED

5. Officer Dashboard

- Summary cards: total projects, active, completed and delayed.
- Land statistics: total parcels, acquired, pending verification and compensation pending.
- Project table with progress, location, department, status and last update.
- Quick links to GIS map, documents, compensation and risk monitoring.
- Charts for progress, pending work and delays.

6. Project Workflow

Project Created → Land Identified → Survey Completed → Documents Submitted → Ownership Verified → Valuation Completed → Compensation Processing → Compensation Approved/Paid → Final Acquisition → Project Completed.

7. GIS Feature

The map is one of the strongest visual features for the SIH demo. Use fictional/demo parcel coordinates and display status by color. Clicking a parcel should open its ID, owner, area, acquisition status and compensation status.

Suggested legend: Green = acquired; Yellow = in progress; Blue = verification pending; Red = delayed/disputed.

8. Intelligent Delay Detection

Start with a transparent rule-based risk engine instead of complicated machine learning. Example: 0–7 days = On Track; 8–15 = Attention; 16–30 = High Risk; 30+ = Critical. The system should explain why an item is flagged and suggest the next action.

Example: Compensation approval pending for 24 days → High Risk → Review pending documents/approval.

9. Landowner Experience

- View only the land parcels associated with the logged-in landowner.
- See acquisition progress as a simple timeline.
- See which documents are verified, pending or rejected.
- See compensation amount and current payment stage.
- Receive notifications when an important status changes.

10. Database — Basic Structure

Recommended entities: users, roles, departments, projects, landowners, land_parcels, documents, document_verifications, project_stages, compensation, notifications, risk_alerts, audit_logs.

Relationships: One project → many land parcels; one landowner → many parcels; one parcel → many documents; one parcel → compensation record; one project → many workflow stages.

11. Suggested Technology Stack

- **Frontend:** React/Next.js + TypeScript + Tailwind CSS.
- **Backend/Database:** Supabase + PostgreSQL is a practical beginner-friendly option.
- **Maps:** Leaflet + OpenStreetMap for the prototype.
- **Charts:** Recharts or a similar chart library.
- **Advanced:** OCR API/integration can be added after the MVP is stable.

12. Six-Member Team Division

Member	Primary responsibility
1	Frontend/UI/UX — layouts, navigation and reusable components
2	Backend/API — business logic and data operations
3	Database/Auth — schema, relationships and role-based access
4	GIS — map, parcels, filters and location visualization
5	Risk/OCR — delay engine first; OCR as advanced feature
6	Dashboard/Integration — analytics, notifications, testing and final integration

13. MVP — What to Build First

Phase 1 (must work): Login → Officer dashboard → Project creation → Land parcels → Documents → Compensation → Landowner portal.

Phase 2: GIS map → notifications → analytics → audit log → search/filtering.

Phase 3: delay/risk engine → OCR → advanced analytics → report export.

Do not attempt every advanced feature before the basic workflow is functional.

14. SIH Demo Story

- Login as a government officer.
- Open an active project and show its progress.
- Open the GIS map and click a parcel.
- Open the parcel's documents and verify one pending document.
- Open compensation and show the status.
- Open the risk monitor and show a delayed project with an explanation.
- Login as the fictional landowner for that parcel.
- Show the same status from the landowner's simpler dashboard.
- Show the notification generated by the status change.
- Finish with the analytics dashboard and explain how the platform improves visibility and monitoring.

15. What Makes the Project Strong

- It solves a connected workflow rather than being only a generic CRUD application.
- GIS gives the project a strong visual demonstration.
- The landowner portal adds transparency and a second user perspective.

- Delay detection provides a measurable intelligent feature.
- Audit logs and role-based access make the prototype feel closer to a real public-sector system.
- The architecture allows OCR/AI features to be added without rebuilding the core system.

16. Avoid These Mistakes

- Do not make many disconnected features that do not share data.
- Do not spend most of the time on animations and visual effects.
- Do not claim real government integration if you only have a prototype.
- Do not use real citizens' personal information in demo data.
- Do not start with ML before the core workflow works.
- Do not build only static screens; important actions should update displayed data.

17. Final Product Definition

BhoomiSetu = Project Management + Land Parcel/GIS + Documents + Verification + Compensation + Landowner Transparency + Delay Monitoring + Analytics.

For a first-year team, the strongest version is not the one with the most features. It is the one where the core workflow is complete, the data is connected, the demo is smooth, and every major feature has a clear purpose.